
Introduction to Database

XXXXXX

Project

Deadline: Monday 09/08/2026 @ 23:59

[Total Mark is 20]

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Instructions:

- You must submit two separate copies (**one Word file and one PDF file**) using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g. misspell words, remove spaces between words, hide characters, use different character sets, **convert text into image** or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You **MUST** show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

Project Instructions

- You can work on this project as a group (minimum 3 and maximum 4 students). Each group member must submit the project individually with all group member names mentioned on the cover page.
- This project **worth 20 marks** and will be distributed as in the following:
 - Database Design. (6 marks)
 - Database Implementation. (6 marks)
 - SQL Operations. (6 marks)
 - Project Reflection. (2 marks)
- Each leader in the group must submit one report about their chosen Project via the Blackboard **(Email submission will not be accepted and will be awarded ZERO marks)** containing the following:
 - A. Task 1 : database design
 - B. Task 2: database implementation
 - C. All requested queries/results
 - D. Screenshots from MySQL (or any other software you use) of **all the tables** after population and **query results**.
 - E. Reflection (300–500 words) describing the team's experience, challenges, design decisions, and AI usage (if applicable).
- Submit one **SQL file** containing the complete database implementation, including database creation, tables, data insertion, required queries, VIEW, and TRIGGER.
- Submit a **GitHub** (or GitLab) repository containing the complete project with regular commits and meaningful commit messages.
- Submit a **progress report** describing the completed work, key decisions, challenges, future plan, and contribution of each group member.
- Include **links** to the live report document, GitHub/GitLab repository, and Google Colab notebook (if applicable) in the **Project Artifacts** section of the final report.
- You are advised to make your work clear and well presented; marks may be reduced for poor presentation. This includes filling in your information on the cover page.
- You **MUST** show all your work, and text **must not** be converted into an image unless specified otherwise by the question.
- **Late submission will result in ZERO marks being awarded.**
- The work should be your own. Copying **from students or other resources, including AI software, will result in ZERO marks.**

*Learning
Outcome(s):*

*CLO1, CLO2, CLO3,
CLO4, CLO5*

Project Description

Design and Implementation of a Smart Clinic Database System

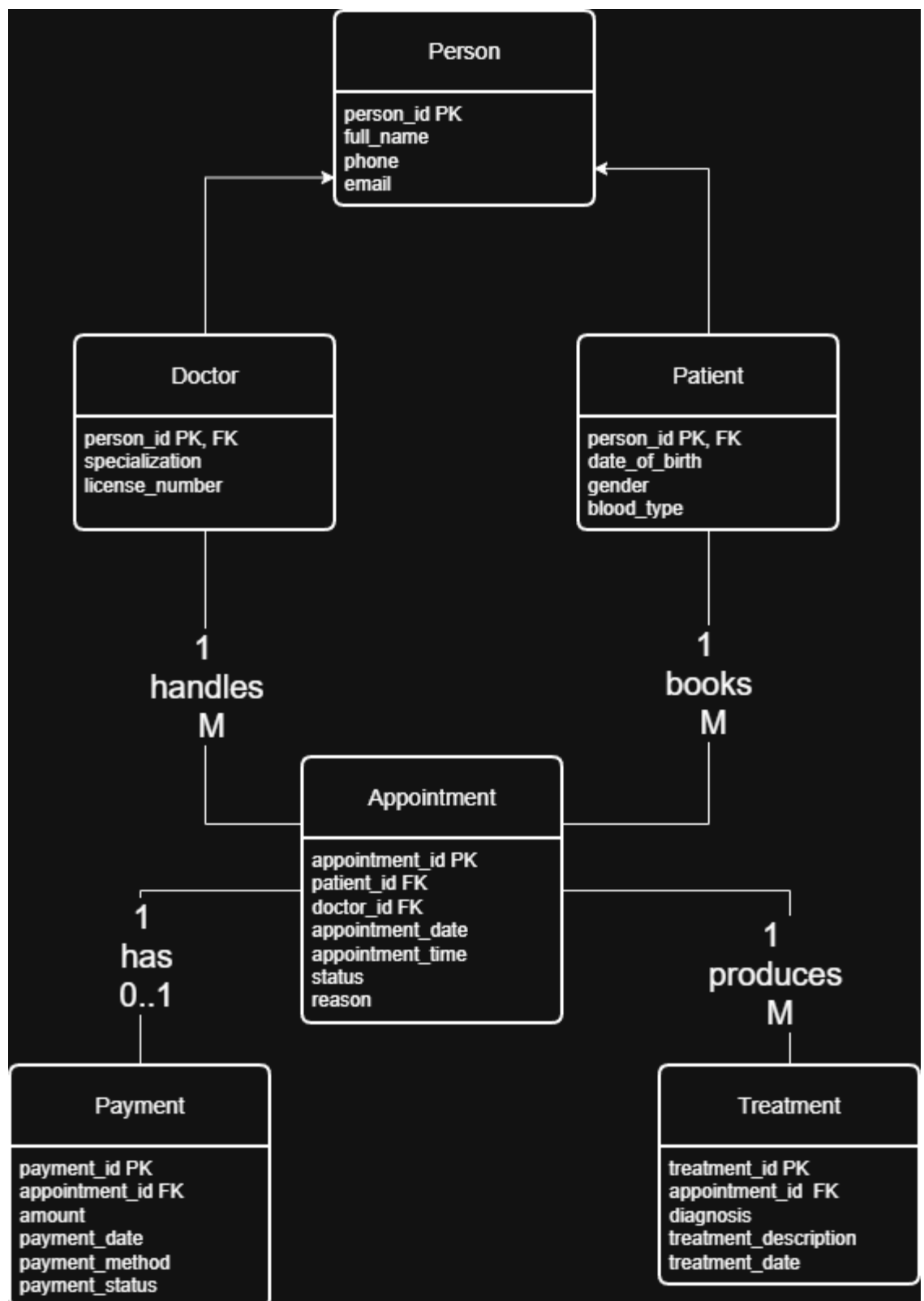
A private clinic currently manages its patient information manually, making it difficult to organize appointments, treatments, and payments. The clinic has decided to develop a database system to improve data organization and retrieval.

Your team is required to design, implement, and test a complete relational database using the concepts learned throughout this course. The project should demonstrate your understanding of database design, ER/EER modeling, relational mapping, SQL implementation, and advanced SQL queries. The project must be completed using MySQL.

Task 1 – Database Design (6 Marks)

- Design a complete database for the clinic that includes at least six entities such as Patients, Doctors, Appointments, Treatments, Medicines, Payments, or any other appropriate entities.
- A complete ER/EER diagram.
- Primary and foreign keys.
- Appropriate relationships with cardinality.
- At least one specialization/generalization (EER feature).
- Any assumptions made during the design process.

Entity	Attributes	Primary key	Foreign Key
Person	full_name, phone, email	person_id	None
Patient	date_of_birth, gender, blood_type	person_id	person_id
Doctor	specialization, license_number	person_id	person_id
Appointment	appointment_date, appointment_time, status, reason	appointment_id	patient_id, doctor_id
Treatment	diagnosis, treatment_description, treatment_date	treatment_id	appointment_id
Payment	amount, payment_date, payment_method, payment_status	payment_id	appointment_id



Note: All Task 1 requirements are included in the table and ER/EER diagram above. The table shows the six entities, their attributes, primary keys, and foreign keys. The diagram shows the relationships, cardinalities, and the specialization of Person into Doctor and Patient. The design assumptions are listed below.

Assumptions:

1. Every patient and doctor is stored as a person.
2. A patient can book many appointments, and a doctor can handle many appointments.
3. Each appointment can have zero or one payment and many treatments.

Task 2 – Database Implementation (6 Marks)

Complete the following tasks using MySQL. Include screenshots of both your SQL code and the corresponding output (results) for each task:

- Create the database and all required tables.
- Use appropriate data types and constraints (PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, etc.).
- Insert at least 5 records into each main table.

```

-- Create database if not exists smartclinic;
-- Use smartclinic;

-- Create table Person
CREATE TABLE Person (
  person_id INT PRIMARY KEY,
  full_name VARCHAR(100) NOT NULL,
  phone VARCHAR(15) NOT NULL UNIQUE,
  email VARCHAR(100) NOT NULL UNIQUE
);

-- Create table Patient
CREATE TABLE Patient (
  person_id INT PRIMARY KEY,
  date_of_birth DATE NOT NULL,
  gender VARCHAR(10) NOT NULL,
  blood_type VARCHAR(10) NOT NULL,
  FOREIGN KEY (person_id) REFERENCES Person(person_id)
);

-- Create table Doctor
CREATE TABLE Doctor (
  person_id INT PRIMARY KEY,
  specialization VARCHAR(100) NOT NULL,
  license_number VARCHAR(50) NOT NULL UNIQUE,
  FOREIGN KEY (person_id) REFERENCES Person(person_id)
);

-- Create table Appointment
CREATE TABLE Appointment (
  appointment_id INT PRIMARY KEY,
  patient_id INT NOT NULL,
  doctor_id INT NOT NULL,
  appointment_date DATE NOT NULL,
  appointment_time TIME NOT NULL,
  reason VARCHAR(200) NOT NULL,
  FOREIGN KEY (patient_id) REFERENCES Patient(person_id),
  FOREIGN KEY (doctor_id) REFERENCES Doctor(person_id)
);

-- Create table Treatment
CREATE TABLE Treatment (
  treatment_id INT PRIMARY KEY,
  appointment_id INT NOT NULL,
  diagnosis VARCHAR(200) NOT NULL,
  treatment_description VARCHAR(200) NOT NULL,
  treatment_date DATE NOT NULL,
  FOREIGN KEY (appointment_id) REFERENCES Appointment(appointment_id)
);

-- Create table Payment
CREATE TABLE Payment (
  payment_id INT PRIMARY KEY,
  appointment_id INT NOT NULL,
  amount DECIMAL(10,2) NOT NULL,
  payment_date DATE NOT NULL,
  payment_method VARCHAR(50) NOT NULL,
  FOREIGN KEY (appointment_id) REFERENCES Appointment(appointment_id)
);

-- Insert into Person table
INSERT INTO Person VALUES
(1, 'Ahmed Ali', '0000000001', 'ahmed@gmail.com'),
(2, 'Sara Mohamed', '0000000002', 'sara@gmail.com'),
(3, 'Omar Ahmed', '0000000003', 'omar@gmail.com'),
(4, 'Nora Salah', '0000000004', 'nora@gmail.com'),
(5, 'Faisal Hassan', '0000000005', 'faisal@gmail.com'),
(6, 'Abdullah Ismail', '0000000006', 'abdullah@gmail.com'),
(7, 'Aman Fahad', '0000000007', 'aman@gmail.com'),
(8, 'Mohamed Hassan', '0000000008', 'mohamed@gmail.com'),
(9, 'Huda Ibrahim', '0000000009', 'huda@gmail.com'),
(10, 'Youssef Rajab', '0000000010', 'youssef@gmail.com');

-- Insert into Patient table
INSERT INTO Patient VALUES
(1, '2000-01-15', 'Male', 'A'),
(2, '1995-03-20', 'Female', 'B'),
(3, '2002-05-10', 'Male', 'O'),
(4, '1998-07-25', 'Female', 'AB'),
(5, '2001-12-01', 'Male', 'A');

-- Insert into Doctor table
INSERT INTO Doctor VALUES
(6, 'General Medicine', 'DX10001'),
(7, 'Cardiology', 'DX10002'),
(8, 'Dermatology', 'DX10003'),
(9, 'Pediatrics', 'DX10004'),
(10, 'Orthopedics', 'DX10005');

-- Insert into Appointment table
INSERT INTO Appointment VALUES
(1, 1, 6, '2024-08-15', '09:00:00', 'Completed', 'General checkup'),
(2, 2, 7, '2024-08-16', '10:00:00', 'Completed', 'Skin allergy'),
(3, 3, 8, '2024-08-17', '11:00:00', 'Completed', 'Regular checkup'),
(4, 4, 9, '2024-08-18', '12:00:00', 'Scheduled', 'Knee pain'),
(5, 5, 10, '2024-08-19', '13:00:00', 'Scheduled', 'Allergy cream');

-- Insert into Treatment table
INSERT INTO Treatment VALUES
(1, 1, 'Common cold', 'Rest and medication', '2024-08-15'),
(2, 2, 'Skin allergy', 'Pain relief medication', '2024-08-16'),
(3, 3, 'Stress allargy', 'Allergy cream', '2024-08-17');

```

Task 2 – Database Implementation (6 Marks)

The screenshot shows the OneCompiler MySQL IDE with two panels. The top panel displays SQL queries to create a database and tables. The bottom panel shows the output of the queries, including the table structure and descriptions.

```

1 CREATE DATABASE IF NOT EXISTS SmartClinic;
2 USE SmartClinic;
3
4 CREATE TABLE Person (
5     person_id INT PRIMARY KEY,
6     full_name VARCHAR(100) NOT NULL,
7     phone VARCHAR(15) NOT NULL UNIQUE,
8     email VARCHAR(100) NOT NULL UNIQUE
9 );
10
11 CREATE TABLE Patient (
12     person_id INT PRIMARY KEY,
13     date_of_birth DATE NOT NULL,
14     gender VARCHAR(10) NOT NULL,
15     blood_type VARCHAR(5) NOT NULL,
16     FOREIGN KEY (person_id) REFERENCES Person(person_id)
17 );
18
19 CREATE TABLE Doctor (
20     person_id INT PRIMARY KEY,
21     specialization VARCHAR(100) NOT NULL,
22     license_number VARCHAR(20) NOT NULL UNIQUE,
23     FOREIGN KEY (person_id) REFERENCES Person(person_id)
24 );
25
26 CREATE TABLE Appointment (
27     appointment_id INT PRIMARY KEY,
28     patient_id INT NOT NULL,
29     doctor_id INT NOT NULL,
30     appointment_date DATE NOT NULL,
31     appointment_time TIME NOT NULL,
32     status VARCHAR(20) NOT NULL,
33     reason VARCHAR(200) NOT NULL,
34     FOREIGN KEY (patient_id) REFERENCES Patient(person_id),
35     FOREIGN KEY (doctor_id) REFERENCES Doctor(person_id)
36 );
37
38 CREATE TABLE Treatment (
39     treatment_id INT PRIMARY KEY,
40     appointment_id INT NOT NULL,
41     diagnosis VARCHAR(200) NOT NULL,
42     treatment_description VARCHAR(200) NOT NULL,
43     treatment_date DATE NOT NULL,
44     FOREIGN KEY (appointment_id) REFERENCES Appointment(appointment_id)
45 );
46
47 CREATE TABLE Payment (
48     payment_id INT PRIMARY KEY,
49     appointment_id INT NOT NULL UNIQUE,
50     amount DECIMAL(10,2) NOT NULL,
51     payment_date DATE NOT NULL,
52     payment_method VARCHAR(20) NOT NULL,
53     payment_status VARCHAR(20) NOT NULL,
54     FOREIGN KEY (appointment_id) REFERENCES Appointment(appointment_id)
55 );
56
57 --
58 --
59 SHOW TABLES;

```

The output shows the table structure for each table, including the field names, data types, and constraints.

Field	Type	Null	Key	Default	Extra
person_id	int	NO	PK		
full_name	varchar(100)	NO			
phone	varchar(15)	NO	UNI		
email	varchar(100)	NO	UNI		

Field	Type	Null	Key	Default	Extra
person_id	int	NO	PK		
date_of_birth	date	NO			
gender	varchar(10)	NO			
blood_type	varchar(5)	NO			

Field	Type	Null	Key	Default	Extra
person_id	int	NO	PK		
specialization	varchar(100)	NO			
license_number	varchar(20)	NO	UNI		

Field	Type	Null	Key	Default	Extra
appointment_id	int	NO	PK		
patient_id	int	NO	FK		
doctor_id	int	NO	FK		
appointment_date	date	NO			
appointment_time	time	NO			
status	varchar(20)	NO			
reason	varchar(200)	NO			

Field	Type	Null	Key	Default	Extra
treatment_id	int	NO	PK		
appointment_id	int	NO	FK		
diagnosis	varchar(200)	NO			
treatment_description	varchar(200)	NO			
treatment_date	date	NO			

Field	Type	Null	Key	Default	Extra
payment_id	int	NO	PK		
appointment_id	int	NO	FK		
amount	decimal(10,2)	NO			
payment_date	date	NO			
payment_method	varchar(20)	NO			
payment_status	varchar(20)	NO			

Task 3 – SQL Operations (6 Marks)

Complete the following tasks using MySQL. For each task, include screenshots of both your SQL code and the corresponding output (results), along with a brief description (one to two sentences) explaining the purpose of the query and the information it is intended to retrieve.

- Write SELECT statements.
- Write JOIN queries.
- Write nested queries.
- Use aggregate functions with GROUP BY.
- Write UPDATE and DELETE statements.
- Create one VIEW.
- Create one TRIGGER.

- **Write SELECT statements:** The first query retrieves all scheduled appointments. The second query retrieves payments with an amount equal to or greater than 200.

- **Write JOIN queries.:** This query joins the Appointment and Person tables. It displays each appointment with the patient's name, doctor's name, date, time, and status.

- **Write nested queries.:** This nested query retrieves the patients who have scheduled appointments.

- **Use aggregate functions with GROUP BY.:** This query groups payments by payment method. It calculates the number of payments and the total amount for each method.

- **Write UPDATE and DELETE statements.:** The UPDATE statement changes appointment number 4 to Completed. The DELETE statement removes payment number 5 from the Payment table.

- **Create one VIEW.:** This view stores appointment details with the patient's name and doctor's name. It can be used to retrieve appointment information easily.

Task 3 – SQL Operations (6 Marks)

- **Create one TRIGGER.:** This trigger automatically sets the status of every new appointment to Scheduled. Appointment number 6 is inserted to demonstrate the trigger.

```

-- Create Table Payment
CREATE TABLE Payment (
    id INT(4) PRIMARY KEY,
    patient_id INT(4),
    doctor_id INT(4),
    appointment_date DATE,
    appointment_time TIME,
    status VARCHAR(20),
    reason VARCHAR(50)
);

-- Insert Data into Person Table
INSERT INTO Person VALUES
(1, 'Ahmed Ali', '0500000001', 'ahmed@gmail.com'),
(2, 'Sara Mohamed', '0500000002', 'sara@gmail.com'),
(3, 'Khalid Omar', '0500000003', 'khalid@gmail.com'),
(4, 'Noura Saleh', '0500000004', 'noura@gmail.com'),
(5, 'Faisal Hassan', '0500000005', 'faisal@gmail.com'),
(6, 'Abdullah Saad', '0500000006', 'abdullah@gmail.com'),
(7, 'Reem Fahad', '0500000007', 'reem@gmail.com'),
(8, 'Mohammed Nasser', '0500000008', 'mohammed@gmail.com'),
(9, 'Huda Ibrahim', '0500000009', 'huda@gmail.com'),
(10, 'Yasser Hajed', '0500000010', 'yasser@gmail.com');

-- Insert Data into Patient Table
INSERT INTO Patient VALUES
(1, '2000-05-15', 'Male', 'A'),
(2, '1998-09-20', 'Female', 'B'),
(3, '2002-01-10', 'Male', 'O'),
(4, '1995-07-25', 'Female', 'M'),
(5, '2001-12-05', 'Male', 'A');

-- Insert Data into Doctor Table
INSERT INTO Doctor VALUES
(6, 'General Medicine', 'DOC1001'),
(7, 'Cardiology', 'DOC1002'),
(8, 'Dermatology', 'DOC1003'),
(9, 'Pediatrics', 'DOC1004'),
(10, 'Orthopedics', 'DOC1005');

-- Insert Data into Appointment Table
INSERT INTO Appointment VALUES
(1, 1, 6, '2020-08-15', '09:00:00', 'Completed', 'General checkup'),
(2, 2, 7, '2020-08-16', '10:00:00', 'Completed', 'Chest pain'),
(3, 3, 8, '2020-08-17', '11:00:00', 'Completed', 'Skin allergy'),
(4, 4, 9, '2020-08-18', '12:00:00', 'Scheduled', 'Regular checkup'),
(5, 5, 10, '2020-08-19', '13:00:00', 'Scheduled', 'Knee pain');

-- Insert Data into Treatment Table
INSERT INTO Treatment VALUES
(1, 1, 'Common cold', 'Rest and medication', '2020-08-15'),
(2, 2, 'Muscle pain', 'Pain relief medication', '2020-08-16'),
(3, 3, 'Skin allergy', 'Allergy cream', '2020-08-17'),
(4, 4, 'Normal condition', 'Regular follow-up', '2020-08-18'),
(5, 5, 'Knee inflammation', 'Rest and physical therapy', '2020-08-19');

-- Insert Data into Payment Table
INSERT INTO Payment VALUES
(1, 1, 150.00, '2020-08-15', 'Cash', 'Paid'),
(2, 2, 300.00, '2020-08-16', 'Card', 'Paid'),
(3, 3, 200.00, '2020-08-17', 'Card', 'Paid'),
(4, 4, 100.00, '2020-08-18', 'Cash', 'Paid'),
(5, 5, 250.00, '2020-08-19', 'Card', 'Paid');

-- Create Trigger SetAppointmentStatus
CREATE TRIGGER SetAppointmentStatus
BEFORE INSERT ON Appointment
FOR EACH ROW
SET NEW.status = 'Scheduled';

-- Insert Appointment 6 to demonstrate the trigger
INSERT INTO Appointment VALUES
(6, 1, 6, '2020-08-20', '14:00:00', 'Pending', 'Follow-up');

-- Select * FROM Appointment
SELECT * FROM Appointment
WHERE appointment_id = 6;

```

appointment_id	patient_id	doctor_id	appointment_date	appointment_time	status	reason
1	1	6	2020-08-15	09:00:00	Completed	General checkup
2	2	7	2020-08-16	10:00:00	Completed	Chest pain
3	3	8	2020-08-17	11:00:00	Completed	Skin allergy
4	4	9	2020-08-18	12:00:00	Scheduled	Regular checkup
5	5	10	2020-08-19	13:00:00	Scheduled	Knee pain
6	1	6	2020-08-20	14:00:00	Pending	Follow-up

```

-- Create View AppointmentDetails AS
SELECT
    a.appointment_id,
    p.full_name AS patient_name,
    d.full_name AS doctor_name,
    a.appointment_date,
    a.appointment_time,
    a.status
FROM Appointment AS a
JOIN Patient AS p
ON a.patient_id = patient.person_id
JOIN Doctor AS d
ON a.doctor_id = doctor.person_id;

-- Select * FROM AppointmentDetails;
SELECT * FROM AppointmentDetails;

```

appointment_id	patient_name	doctor_name	appointment_date	appointment_time	status
1	Ahmed Ali	Abdullah Saad	2020-08-15	09:00:00	Completed
2	Sara Mohamed	Reem Fahad	2020-08-16	10:00:00	Completed
3	Khalid Omar	Mohammed Nasser	2020-08-17	11:00:00	Completed
4	Noura Saleh	Huda Ibrahim	2020-08-18	12:00:00	Scheduled
5	Faisal Hassan	Yasser Hajed	2020-08-19	13:00:00	Scheduled

Task 3 – SQL Operations (6 Marks)

```
queries.sql X +
15;
16;
17 INSERT INTO Person VALUES
18 (1, 'Ahmed Ali', '0500000001', 'ahmed@gmail.com'),
19 (2, 'Sara Mohammed', '0500000002', 'sara@gmail.com'),
20 (3, 'Khalid Omar', '0500000003', 'khalid@gmail.com'),
21 (4, 'Noura Saleh', '0500000004', 'noura@gmail.com'),
22 (5, 'Faisal Hassan', '0500000005', 'faisal@gmail.com'),
23 (6, 'Abdullah Saad', '0500000006', 'abdullah@gmail.com'),
24 (7, 'Reem Fahad', '0500000007', 'reem@gmail.com'),
25 (8, 'Mohammed Hassan', '0500000008', 'mohammed@gmail.com'),
26 (9, 'Huda Ibrahim', '0500000009', 'huda@gmail.com'),
27 (10, 'Yasser Majed', '0500000010', 'yasser@gmail.com');
28;
29 INSERT INTO Patient VALUES
30 (1, '2000-05-15', 'Male', 'A'),
31 (2, '1998-09-20', 'Female', 'B'),
32 (3, '2002-01-18', 'Male', 'O+'),
33 (4, '1995-07-25', 'Female', 'AB+'),
34 (5, '2001-12-05', 'Male', 'A-');
35;
36 INSERT INTO Doctor VALUES
37 (6, 'General Medicine', 'DOC1001'),
38 (7, 'Cardiology', 'DOC1002'),
39 (8, 'Dermatology', 'DOC1003'),
40 (9, 'Pediatrics', 'DOC1004'),
41 (10, 'Orthopedics', 'DOC1005');
42;
43 INSERT INTO Appointment VALUES
44 (1, 1, 6, '2026-08-15', '09:00:00', 'Completed', 'General checkup'),
45 (2, 2, 7, '2026-08-16', '10:00:00', 'Completed', 'Chest pain'),
46 (3, 3, 8, '2026-08-17', '11:00:00', 'Completed', 'Skin allergy'),
47 (4, 4, 9, '2026-08-18', '12:00:00', 'Scheduled', 'Regular checkup'),
48 (5, 5, 10, '2026-08-19', '13:00:00', 'Scheduled', 'Knee pain');
49;
50 INSERT INTO Treatment VALUES
51 (1, 1, 'Common cold', 'Rest and medication', '2026-08-15'),
52 (2, 2, 'Muscle pain', 'Pain relief medication', '2026-08-16'),
53 (3, 3, 'Skin allergy', 'Allergy cream', '2026-08-17'),
54 (4, 4, 'Normal condition', 'Regular follow-up', '2026-08-18'),
55 (5, 5, 'Knee inflammation', 'Rest and physical therapy', '2026-08-19');
56;
57 INSERT INTO Payment VALUES
58 (1, 1, 150.00, '2026-08-15', 'Cash', 'Paid'),
59 (2, 2, 300.00, '2026-08-16', 'Card', 'Paid'),
60 (3, 3, 200.00, '2026-08-17', 'Card', 'Paid'),
61 (4, 4, 100.00, '2026-08-18', 'Cash', 'Paid'),
62 (5, 5, 250.00, '2026-08-19', 'Card', 'Paid');
63;
64 UPDATE Appointment
65 SET status = 'Completed'
66 WHERE appointment_id = 4;
67;
68 SELECT * FROM Appointment
69 WHERE appointment_id = 4;
70;
71 DELETE FROM Payment
72 WHERE payment_id = 5;
73;
74 SELECT * FROM Payment;
```

```
queries.sql X +
67;
68 CREATE TABLE Payment (
69     payment_date DATE NOT NULL,
70     payment_method VARCHAR(30) NOT NULL,
71     payment_status VARCHAR(20) NOT NULL,
72     FOREIGN KEY (appointment_id) REFERENCES Appointment(appointment_id)
73 );
74;
75 INSERT INTO Person VALUES
76 (1, 'Ahmed Ali', '0500000001', 'ahmed@gmail.com'),
77 (2, 'Sara Mohammed', '0500000002', 'sara@gmail.com'),
78 (3, 'Khalid Omar', '0500000003', 'khalid@gmail.com'),
79 (4, 'Noura Saleh', '0500000004', 'noura@gmail.com'),
80 (5, 'Faisal Hassan', '0500000005', 'faisal@gmail.com'),
81 (6, 'Abdullah Saad', '0500000006', 'abdullah@gmail.com'),
82 (7, 'Reem Fahad', '0500000007', 'reem@gmail.com'),
83 (8, 'Mohammed Hassan', '0500000008', 'mohammed@gmail.com'),
84 (9, 'Huda Ibrahim', '0500000009', 'huda@gmail.com'),
85 (10, 'Yasser Majed', '0500000010', 'yasser@gmail.com');
86;
87 INSERT INTO Patient VALUES
88 (1, '2000-05-15', 'Male', 'A'),
89 (2, '1998-09-20', 'Female', 'B'),
90 (3, '2002-01-18', 'Male', 'O+'),
91 (4, '1995-07-25', 'Female', 'AB+'),
92 (5, '2001-12-05', 'Male', 'A-');
93;
94 INSERT INTO Doctor VALUES
95 (6, 'General Medicine', 'DOC1001'),
96 (7, 'Cardiology', 'DOC1002'),
97 (8, 'Dermatology', 'DOC1003'),
98 (9, 'Pediatrics', 'DOC1004'),
99 (10, 'Orthopedics', 'DOC1005');
100;
101 INSERT INTO Appointment VALUES
102 (1, 1, 6, '2026-08-15', '09:00:00', 'Completed', 'General checkup'),
103 (2, 2, 7, '2026-08-16', '10:00:00', 'Completed', 'Chest pain'),
104 (3, 3, 8, '2026-08-17', '11:00:00', 'Completed', 'Skin allergy'),
105 (4, 4, 9, '2026-08-18', '12:00:00', 'Scheduled', 'Regular checkup'),
106 (5, 5, 10, '2026-08-19', '13:00:00', 'Scheduled', 'Knee pain');
107;
108 INSERT INTO Treatment VALUES
109 (1, 1, 'Common cold', 'Rest and medication', '2026-08-15'),
110 (2, 2, 'Muscle pain', 'Pain relief medication', '2026-08-16'),
111 (3, 3, 'Skin allergy', 'Allergy cream', '2026-08-17'),
112 (4, 4, 'Normal condition', 'Regular follow-up', '2026-08-18'),
113 (5, 5, 'Knee inflammation', 'Rest and physical therapy', '2026-08-19');
114;
115 INSERT INTO Payment VALUES
116 (1, 1, 150.00, '2026-08-15', 'Cash', 'Paid'),
117 (2, 2, 300.00, '2026-08-16', 'Card', 'Paid'),
118 (3, 3, 200.00, '2026-08-17', 'Card', 'Paid'),
119 (4, 4, 100.00, '2026-08-18', 'Cash', 'Paid'),
120 (5, 5, 250.00, '2026-08-19', 'Card', 'Paid');
121;
122 SELECT
123     payment_method,
124     COUNT(*) AS total_payments,
125     SUM(amount) AS total_amount
126 FROM Payment
127 GROUP BY payment_method;
```

```
queries.sql X +
125;
126;
127;
128;
129;
130;
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Task 3 – SQL Operations (6 Marks)

Query 1: Create Table Payment

```

CREATE TABLE Payment (
    payment_id VARCHAR(10) NOT NULL,
    payment_status VARCHAR(20) NOT NULL,
    FOREIGN KEY (appointment_id) REFERENCES Appointment(appointment_id)
);

INSERT INTO Person VALUES
(1, 'Ahmed Ali', '0500000001', 'ahmed@gmail.com'),
(2, 'Sara Mohamed', '0500000002', 'sara@gmail.com'),
(3, 'Walid Omar', '0500000003', 'walid@gmail.com'),
(4, 'Noura Salah', '0500000004', 'noura@gmail.com'),
(5, 'Faisal Hassan', '0500000005', 'faisal@gmail.com'),
(6, 'Abdullah Saad', '0500000006', 'abdullah@gmail.com'),
(7, 'Rana Fahad', '0500000007', 'rana@gmail.com'),
(8, 'Muhammad Bassem', '0500000008', 'muhammad@gmail.com'),
(9, 'Huda Ibrahim', '0500000009', 'huda@gmail.com'),
(10, 'Yasser Hajed', '0500000010', 'yasser@gmail.com');

INSERT INTO Patient VALUES
(1, '2000-05-15', 'Male', 'M'),
(2, '1990-03-20', 'Female', 'F'),
(3, '2002-01-10', 'Male', 'M'),
(4, '1995-07-25', 'Female', 'F'),
(5, '2001-12-05', 'Male', 'M');

INSERT INTO Doctor VALUES
(6, 'General Medicine', 'DOC1001'),
(7, 'Cardiology', 'DOC1002'),
(8, 'Dermatology', 'DOC1003'),
(9, 'Pediatrics', 'DOC1004'),
(10, 'Orthopedics', 'DOC1005');

INSERT INTO Appointment VALUES
(1, 1, 6, '2020-08-15', '09:00:00', 'Completed', 'General checkup'),
(2, 2, 7, '2020-08-15', '10:00:00', 'Completed', 'Chest pain'),
(3, 3, 8, '2020-08-17', '11:00:00', 'Completed', 'Skin allergy'),
(4, 4, 9, '2020-08-18', '12:00:00', 'Scheduled', 'Regular checkup'),
(5, 5, 10, '2020-08-19', '13:00:00', 'Scheduled', 'Knee pain');

INSERT INTO Treatment VALUES
(1, 1, 'Common cold', 'Rest and medication', '2020-08-15'),
(2, 2, 'Muscle pain', 'Pain relief medication', '2020-08-16'),
(3, 3, 'Skin allergy', 'Allergy cream', '2020-08-17'),
(4, 4, 'Nasal congestion', 'Regular follow-up', '2020-08-18'),
(5, 5, 'Knee inflammation', 'Rest and physical therapy', '2020-08-19');

INSERT INTO Payment VALUES
(1, 1, 100.00, '2020-08-15', 'Cash', 'Paid'),
(2, 2, 200.00, '2020-08-16', 'Card', 'Paid'),
(3, 3, 200.00, '2020-08-17', 'Card', 'Paid'),
(4, 4, 100.00, '2020-08-18', 'Cash', 'Paid'),
(5, 5, 250.00, '2020-08-19', 'Card', 'Paid');

FROM Person
WHERE person_id IN (
    SELECT patient_id
    FROM Appointment
    WHERE status = 'Scheduled'
);

```

Query 2: Select Appointment Details

```

SELECT
    a.appointment_id,
    p.full_name AS patient_name,
    d.full_name AS doctor_name,
    a.appointment_date,
    a.appointment_time,
    a.status
FROM Appointment AS a
JOIN Person AS patient
ON a.patient_id = patient.person_id
JOIN Doctor AS doctor
ON a.doctor_id = doctor.person_id;

```

Output:

person_id	full_name
4	Noura Salah
5	Faisal Hassan

appointment_id	patient_name	doctor_name	appointment_date	appointment_time	status
1	Ahmed Ali	Abdullah Saad	2020-08-15	09:00:00	Completed
2	Sara Mohamed	Rana Fahad	2020-08-16	10:00:00	Completed
3	Walid Omar	Muhammad Bassem	2020-08-17	11:00:00	Completed
4	Noura Salah	Huda Ibrahim	2020-08-18	12:00:00	Scheduled
5	Faisal Hassan	Yasser Hajed	2020-08-19	13:00:00	Scheduled

```
queries.sql X +
12 CREATE TABLE Payment (
13     payment_method VARCHAR(40) NOT NULL,
14     payment_status VARCHAR(20) NOT NULL,
15     FOREIGN KEY (appointment_id) REFERENCES Appointment(appointment_id)
16 );
17 INSERT INTO Person VALUES
18 (1, 'Ahmed Ali', '0500000001', 'ahmed@gmail.com'),
19 (2, 'Sara Mohamed', '0500000002', 'sara@gmail.com'),
20 (3, 'Khalid Omar', '0500000003', 'khalid@gmail.com'),
21 (4, 'Moura Saleh', '0500000004', 'moura@gmail.com'),
22 (5, 'Faisal Hassan', '0500000005', 'faisal@gmail.com'),
23 (6, 'Abdullah Saad', '0500000006', 'abdullah@gmail.com'),
24 (7, 'Reem Fahad', '0500000007', 'reem@gmail.com'),
25 (8, 'Mohammed Nasser', '0500000008', 'mohammed@gmail.com'),
26 (9, 'Huda Ibrahim', '0500000009', 'huda@gmail.com'),
27 (10, 'Yasser Majed', '0500000010', 'yasser@gmail.com');
28
29 INSERT INTO Patient VALUES
30 (1, '2000-01-15', 'Male', 'A'),
31 (2, '1998-09-20', 'Female', 'B'),
32 (3, '2002-01-10', 'Male', 'C'),
33 (4, '1995-07-25', 'Female', 'A'),
34 (5, '2001-12-05', 'Male', 'A');
35
36 INSERT INTO Doctor VALUES
37 (6, 'General Medicine', 'DOC1001'),
38 (7, 'Cardiology', 'DOC1002'),
39 (8, 'Dermatology', 'DOC1003'),
40 (9, 'Pediatrics', 'DOC1004'),
41 (10, 'Orthopedics', 'DOC1005');
42
43 INSERT INTO Appointment VALUES
44 (1, 1, 6, '2026-08-15', '09:00:00', 'Completed', 'General checkup'),
45 (2, 2, 7, '2026-08-16', '10:00:00', 'Completed', 'Chest pain'),
46 (3, 3, 8, '2026-08-17', '11:00:00', 'Completed', 'Skin allergy'),
47 (4, 4, 9, '2026-08-18', '12:00:00', 'Scheduled', 'Regular checkup'),
48 (5, 5, 10, '2026-08-19', '13:00:00', 'Scheduled', 'Knee pain');
49
50 INSERT INTO Treatment VALUES
51 (1, 1, 'Common cold', 'Rest and medication', '2026-08-15'),
52 (2, 2, 'Muscle pain', 'Pain relief medication', '2026-08-16'),
53 (3, 3, 'Skin allergy', 'Allergy cream', '2026-08-17'),
54 (4, 4, 'Normal condition', 'Regular follow-up', '2026-08-18'),
55 (5, 5, 'Knee inflammation', 'Rest and physical therapy', '2026-08-19');
56
57 INSERT INTO Payment VALUES
58 (1, 1, 150.00, '2026-08-15', 'Cash', 'Paid'),
59 (2, 2, 300.00, '2026-08-16', 'Card', 'Paid'),
60 (3, 3, 200.00, '2026-08-17', 'Card', 'Paid'),
61 (4, 4, 100.00, '2026-08-18', 'Cash', 'Paid'),
62 (5, 5, 250.00, '2026-08-19', 'Card', 'Paid');
63
64 SELECT appointment_id, appointment_date, appointment_time, status
65 FROM Appointment
66 WHERE status = 'Scheduled';
67
68 SELECT payment_id, amount, payment_method, payment_status
69 FROM Payment
70 WHERE amount >= 200;
```

SQL AI Agent

Input for the program (optional)

Output:

appointment_id	appointment_date	appointment_time	status
4	2026-08-18	12:00:00	Scheduled
5	2026-08-19	13:00:00	Scheduled

payment_id	amount	payment_method	payment_status
2	300.00	Card	Paid
3	200.00	Card	Paid
5	250.00	Card	Paid

Task 4 – Reflection (2 Marks)

Write a **300–500-word reflection** on your experience while completing this project by answering the following questions. Your responses should be clear, concise, and supported with specific examples from your own work.

1. Describe one specific challenge or obstacle you encountered during this project and explain in detail how you identified and resolved it.
2. Explain why you chose your specific approach/method/model over at least one alternative you considered, and what tradeoffs informed that decision.
3. If you used any AI tools during this project (e.g., ChatGPT, Claude, Copilot), disclose which tool(s), for what specific purpose (e.g., debugging, grammar checking, brainstorming), and how you verified or modified the output.
4. What would you do differently if you were to redo this project with more time or resources?

During this project, I faced some challenges while connecting the tables and deciding where to place the primary and foreign keys. I solved these issues by reviewing the database design, checking the relationships, and testing the tables in MySQL. Running the SQL statements helped me identify problems and correct them before completing the final report.

I chose a relational database because the clinic information is connected. Patients book appointments, doctors handle appointments, and appointments can have treatments and payments. I used Person as a general entity for Patient and Doctor because they share basic information such as name, phone number, and email. Another possible approach was to store this information separately in Patient and Doctor, but that would repeat the same attributes. The selected approach reduces repetition, although it sometimes requires JOIN queries to retrieve complete information.

I used ChatGPT to understand some project instructions, organize the report, improve the grammar, and review the clarity of my explanations. I checked its suggestions and changed the wording when necessary to make sure the final report matched the project requirements.

If I completed the project again, I would organize the work earlier and test each part immediately after completing it. I would also prepare the table design before creating the ER/EER

diagram so that the attributes and relationships remain consistent. With more time, I would add more sample records and test additional situations, such as duplicate values and incorrect foreign keys. I would also improve the organization of the screenshots to make the report clearer and easier to follow.

Deliverables

1. **Project Report (PDF)** – System description, ER/EER diagram, relational schema, SQL code, screenshots, and reflection.
2. **SQL Script (.sql)** – Complete SQL implementation of the project.
3. **GitHub Repository** – Repository link with the complete project and commit history.
4. **Mid-Project Progress Report** – Summary of progress, challenges, decisions, and team contributions.
5. **Project Artifacts**
 - Live report document (with edit history): [link]
 - GitHub/GitLab repository (with commit history): [[link](#)]
 - Google Colab notebook (if applicable): [link]